Deevi Rahul Bharadwaz

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GitHub **LinkedIn** **LeetCode** **Codeforces**

Education

Indian Institute of Information Technology, Lucknow *Lucknow, India*
Bachelor of Technology in Computer Science and Artificial Intelligence, CGPA: 8.14/10 *Aug 2023 – May 2027*

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming (OOP) & Design Patterns, Database Management Systems (DBMS), Operating Systems, System Design (LLD), Probability & Statistics, Computer Graphics

Technical Skills

AI/ML & Deep Learning: Machine Learning, Deep Learning, NLP, PyTorch, Scikit-learn, Sentence Transformers, LightGBM
Languages: Python, C++, JavaScript, TypeScript, SQL
Systems & Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Real-Time Systems
Backend & Tools: FastAPI, Node.js, Express.js, MongoDB, REST APIs, React, Tailwind CSS, Git

Projects

- **UniLoop – Peer-to-Peer Campus Marketplace** GitHub
  Tech Stack: React, Node.js, Express.js, MongoDB, Socket.IO, Tailwind CSS
  - Designed a multi-tenant platform across **100+ universities**, enforcing campus-scoped data isolation with JWT access/refresh token rotation and secure Resend API email verification.
  - Implemented a WebSocket synchronization layer, maintaining sub-50ms event propagation latency, and structured an 8-stage automated rental workflow using background schedulers.
  - Improved application performance using Zustand-based state management, optimistic updates, Vite code splitting across **4 vendor modules**, and Vercel edge caching, achieving sub-second LCP.
- **CampusLink – Scalable P2P Video Platform** GitHub
  Tech Stack: React, FastAPI, Node.js, MongoDB, Redis, WebRTC, Socket.IO
  - Built a fault-tolerant video engine with an automated 12-second connection watchdog, enabling seamless fallback from P2P WebRTC to cloud servers upon ICE negotiation failure.
  - Optimized matching latency by **70%** via tuned polling loops and engineered an in-browser telemetry suite to parse SDP candidates and diagnose NAT traversal issues.
  - Developed a cross-device real-time chat interface and enhanced render performance through on-demand code splitting of heavy media SDKs.
- **InternMatch – AI Internship Recommendation Platform** GitHub
  Tech Stack: Python, PyTorch, Sentence Transformers, Scikit-learn, Flask, SQLite
  - Engineered an automated parsing engine for PDF/DOCX resumes with regex PII redaction and a 6-category canonical skill taxonomy, mapping candidates to opportunities via 384-dimensional dense Sentence-BERT (`all-MiniLM-L6-v2`) embeddings.
  - Developed an explainable hybrid ranker combining SBERT cosine similarity, TF-IDF lexical matching, and skill overlap with in-memory caching (<5ms latency); benchmarked on 5 personas achieving **0.8667 P@3** and **0.9572 NDCG@3** across 55 pytest tests.

Achievements

- **Amazon ML Challenge 2026 – Business Entity Resolution at Scale:** Ranked **923** as **Team Leader** [GitHub]; led a 3-member team developing an entity matching pipeline using TF-IDF retrieval, a PyTorch Siamese encoder with InfoNCE contrastive learning, and LightGBM reranking.
- **Data Structures & Algorithms:** Solved **800+ problems** across LeetCode, Codeforces, and CodeChef, focusing on optimizing **time and space complexity** and handling edge cases.
- **Analytical Proficiency:** Ranked as **Knight (1806 Rating)** on LeetCode, demonstrating strong algorithmic reasoning and logic under time constraints.
- **Competitive Programming:** Achieved **Specialist** on Codeforces and **3-Star Coder** on CodeChef.
- **National Competitions:** Recognized as a **National Semi-Finalist** in **Flipkart GRID 7.0**.
- **Performance Engineering:** Authored an IEEE-formatted research paper on JSON compression, analyzing data payloads to achieve a **92.4% reduction** via Brotli and Gzip.
- Secured admission to **IIIT Lucknow** by ranking in the **top 1.3%** of 1.2M+ candidates in **JEE Mains 2023**.

Positions of Responsibility

- **PR & Tech Writer** Mar 2024 – May 2024
  Equinox (Tech-Cultural Fest)
  - Managed technical communications and public relations for a **1,000+ attendee** festival, improving community engagement by **40%**.